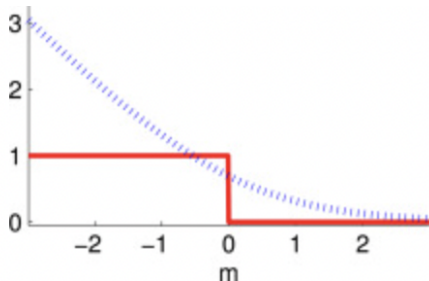


Introduction to KNN

Procheta Sen



World's simplest classifier!

- ▶ Given a training dataset

$$D_{\text{train}}$$

of N instances $(\mathbf{X}, y)$, simply *remember* the entire N instances in memory (or hard-disk): **memory-based learning**.

- ▶ When we want to classify a test (previously unseen, not in D_{train}) instance $\mathbf{X}'$, we would simply check whether $\mathbf{X}'$ is in D_{train} .
- ▶ If $\mathbf{X}' \in D_{\text{train}}$, then return the label of $\mathbf{X}'$.
- ▶ Otherwise, make a random guess.

The Nearest Neighbour classifier

- ▶ **Training:** store entire training set D_{train} .
- ▶ **Classification:** for an input object $\mathbf{X}'$, find in the training set the *closest* object $\mathbf{X}$ to $\mathbf{X}'$ and classify $\mathbf{X}'$ the same as $\mathbf{X}$.

The k -Nearest Neighbour classifier

- ▶ **Training:** store entire training set.
- ▶ **Classification:** for an input object $\mathbf{X}'$,
 - ▶ find in the training set the k closest (nearest) objects to $\mathbf{X}'$,
 - ▶ find the majority label among those nearest neighbours,
 - ▶ the majority label is predicted for the test instance $\mathbf{X}'$.

Example

5-NN: find 5 closest to the black point. 3 blue and 2 red, so predict blue

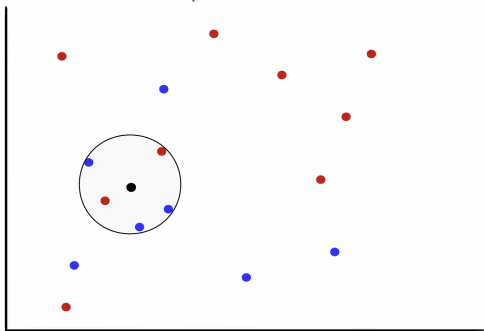


Figure: Caption